

Project: BREATHEBEAT

Team Members:

Sri Harshit K – 2024102022 — Parth Dhodapkar – 2024111009 — Atharva Kulkarni – 2023101072

1. Motivation & Context

Intense mental work quickly burns through our attention spans and slows our reaction times. Most people deal with this by taking short screen breaks or resting quietly, but these resets often fail to actually restore cognitive clarity. While guided breathing and mindfulness apps attempt to solve this, their repetitive synthetic voices and robotic counts can easily become another cognitive distraction.

BREATHEBEAT explores a more intuitive alternative: mapping structured breathing patterns onto melodic audio synthesized in real time. Rooted in Indian classical music, the system translates paced diaphragmatic breathing directly into the organic glides of Raga Bhupali. By pairing this engine with controlled cognitive tests, we want to see whether melodic audio helps students bounce back from mental fatigue faster and more naturally than conventional voice prompts or silent rest.

2. Study Setup & Workflow

We are evaluating the system with a cohort of 20 students through a guided, web-based session. To track performance changes cleanly, the experiment runs in five straightforward stages:

- **1. Baseline Check (2 min):** A standard Stroop test to log each participant’s fresh response time and accuracy.
- **2. Fatigue Inducer (3 min):** Rapid, timed arithmetic and logic puzzles designed to drain working memory and induce focus fatigue.
- **3. Post-Fatigue Check (2 min):** A second Stroop test to record the drop in mental processing speed and precision.
- **4. Guided Rest Break (5 min):** Participants close their eyes and complete a 4-2-6 breathing cycle under one of three randomly assigned audio conditions.
- **5. Recovery Check (2 min):** A final Stroop test to measure how much cognitive capacity returned following the break.

To prevent familiarity from skewing the results, test variants are counterbalanced across steps, and each student is assigned to only one audio condition.

3. The Audio Experience & Musical Design

During the 5-minute break, participants practice 4-2-6 diaphragmatic breathing (4s inhale, 2s hold, 6s exhale). We compare three arms to test our hypothesis:

3.1 Three Comparison Arms

- **Silent Rest (Control):** Quiet rest with gentle chimes marking the beginning and end of the session.
- **Spoken Guide (Baseline):** A calm, synthetic voice counting out pace markers (“*Inhale 2, 3, 4; Hold 2; Exhale 2, 3, 4, 5, 6*”).
- **BREATHEBEAT Engine (Experimental):** Fluid melodic glides in **Raga Bhupali** set over a warm continuous *Tanpura* drone (*Sa-Pa*).

3.2 Direct Audio-to-Breath Mapping

Rather than playing a static melody over breathing instructions, the music sonifies the physical act of breathing itself:

- **Inhale (4s – *Arohana*):** An ascending pitch sweep ($Sa \rightarrow Re \rightarrow Ga \rightarrow Pa \rightarrow Dha \rightarrow Sa'$) with swelling volume to signal air intake naturally.
- **Hold (2s – *Kumbhaka*):** Pitch pauses and sustains steadily on the dominant root note (Ga) over the drone, anchoring breath retention without tension.
- **Exhale (6s – *Avarohana*):** A gentle descending glide ($Sa' \rightarrow Dha \rightarrow Pa \rightarrow Ga \rightarrow Re \rightarrow Sa$) tapering in volume to prompt complete, relaxed release.

3.3 Why Raga Bhupali Works

We chose Raga Bhupali because its technical anatomy aligns with what a fatigued nervous system needs:

- **Clean Pentatonic Scale (*Audava*):** By using only five notes (Sa, Re, Ga, Pa, Dha) and skipping tension-bearing tones like Ma and Ni , the raga avoids unresolved dissonance. Every note harmonizes smoothly with the drone.
- **Mirrored Contours:** The ascent and descent mirror each other identically, creating a smooth arc that naturally mimics the expansion and contraction of the chest.
- **Natural Resting Note:** The raga’s primary emphasis note (*Vadi*) is Ga , which matches the pitch held during the breath retention phase, providing a stable auditory center.
- **Naturally Calming Character:** Traditionally associated with calm, reflective, and meditative settings, Bhupali avoids sharp or agitated melodic turns that could raise cognitive arousal.

4. Metrics & Analysis

To keep our findings clear and actionable, we narrowed the study to three core indicators: cognitive restoration, user cognitive effort, and baseline familiarity.

Metric	What It Tells Us
1. Cognitive Recovery Score	Shows how much of a participant’s lost focus and processing speed was restored after the break, derived from pre- and post-break Stroop reaction times (RT in ms) and accuracy (%). Formulated as: $\text{Recovery} = \frac{X_{\text{fatigue}} - X_{\text{recovery}}}{X_{\text{fatigue}} - X_{\text{baseline}}} \times 100\%$ A score near 100% reflects a complete return to baseline capability, while lower or negative values reveal persistent mental exhaustion.
2. Perceived Effort	Measures how mentally demanding or distracting the breathing condition felt, captured via a quick post-session NASA-TLX ease/effort survey. Tells us whether raga audio allows participants to relax without working hard to track counts.
3. Musical Familiarity	A brief intake rating (1–5) of prior exposure to Indian classical music, helping us verify whether the calming impact is universal or depends on cultural familiarity.

Table 1: Consolidated Evaluation Metrics

Analysis Plan: We will compare Cognitive Recovery and Effort scores across the three experimental groups using a one-way ANOVA (or a non-parametric equivalent if variance is skewed), with pairwise post-hoc tests to determine whether melodic audio yields a statistically meaningful advantage over silence or voice prompts.

5. Deliverables

The project produces a self-contained web platform that runs directly in any modern desktop browser:

1. **Automated Testing Module:** Delivers randomized Stroop tasks and timed arithmetic challenges while recording millisecond-accurate timing and error logs.
2. **Real-Time Web Audio Synthesizer:** Uses the Web Audio API to generate continuous, artifact-free pitch sweeps (*Meend*) and drone sounds timed to custom breath ratios.
3. **Data Dashboard:** Exports trial data and automatically renders clean recovery comparisons across the three test conditions.

6. Significance

Most wellness tools rely on subjective user feedback to claim they reduce stress. By keeping breathing pace and posture identical across all test groups, this project evaluates whether the melodic architecture of Indian classical music offers an objective, measurable boost to cognitive recovery over silence and vocal pacing.